

Communication and Internet Technology: Address Resolution Between IPv4 and IPv6, and Mobile IP

IPv4 address is a stateful address. This means that, if a node moves from one subnet to another, the user has to either reconfigure the IP address or request for a new IP address from dynamic host configuration protocol. Whereas, IPv6 supports a stateless auto configuration address. Thus, with IPv6, while moving from a subnet to another, a host can generate its own IP address.

I Pv4 and IPv6 address spaces are of different sizes. Address notations for the two are also different. IPv6 address of 128 bits is represented by hexadecimal colon notation. 128 bits are divided into eight segments, each two bytes in length. Each of the segments is separated by a colon. As an example, EFEF:1234:DCEE:7676:1C2C:00FE:AB00:D8B9 represents one IPv6 address. The notion allows to drop leading zeros. As a case, in the example, 00FE can be just FE. Moreover, IPv6 frame provides many flexibilities and options over IPv4 frame. One example of IPv4 address representation of 32 bits is 191.245.245.245, a representation of dot-separated four bytes.

Deployment of IPv6 therefore requires some techniques of compatibility to make IPv6 co-exist and co-operate with the existing vast IPv4 network. After all, it is neither technically-feasible nor economically-viable to upgrade all the existing millions of machines of IPv4 to IPv6.

Two schemes used for compatibility are dual stack and tunnelling. Dual stack offers a technique where all nodes in any network are both IPv4- and IPv6-enabled. While deploying, IPv6 nodes can be made backwards-compatible, making them capable of sending IPv4 datagram, too.

However, already existing IPv4 nodes are not capable of sending IPv6 datagram. In dual stack, each node is made IPv6/IPv4-compatible. IPv6 node is made to implement complete IPv4 as well. The node also has the capability to determine whether the next node of routing is IPv4 or IPv6. This is done by the domain name system (DNS). Until and unless the entire world switches to IPv6, dual stack will remain one of the top transition methods for address resolution and a solution for resolving incompatibilities.

In dual stack, any node of IPv6 duly sends IPv6 datagram to the next node, if that node is of IPv6. But if the next node is of IPv4, the received IPv6 datagram is converted to IPv4 datagram before transmitting. By the process

of conversion, a lot of information of the original IPv6 datagram like that of flow label, etc, is lost. This is a major disadvantage of dual stack.

Even if the next-to-next connected node is of IPv6, under dual stack, as IPv6 is equipped to handle IPv4, datagram will flow as IPv4 datagram and not as original IPv6 datagram. The problem with dual stack may be resolved with tunnelling. Tunnelling allows to run IPv6 over IPv4 without loss of any information that IPv6 carries.

In tunnelling, any node, when transmitting IPv6 datagram to IPv4 node, treats the whole IPv6 datagram as payload. With this payload, a new IPv4 datagram is made. IPv4 nodes treat all IPv6 packets as complete payloads for subsequent transmission in tunnelling. In the stated process, information of IPv6 datagram is preserved, but overhead increases. As a result, coding efficiency of the datagram, so framed, decreases. Thus, trade-off between dual stack and tunnelling is loss of information versus decreased coding efficiency.